Justification for Accelerated Publication

The tetrad (vierbein) has been the standard variable for coupling gravity to fermionic matter since the work of Cartan, Einstein, and Weyl. It underpins Poincaré gauge gravity, Ashtekar variables, loop quantum gravity, bimetric gravity, and supergravity. Despite over a century of use, the coordinate-space geometry of the tetrad's spatial sector has not, to our knowledge, been derived in coordinate space.

This letter identifies a previously unreported local geometric structure of the tetrad frame. The spatial sector of a single orthonormal tetrad, expressed as four oriented half-space inequalities, yields dual interlocking tetrahedra: Kepler's stella octangula (1619), here derived within the tetrad frame for the first time. All structural constants of the resulting geometry reduce to two integers: the number of tetrahedral faces and the spatial dimension. No dual frame is introduced; the orientation-reversed tetrahedron emerges as the unique second bounded region of the same construction. The result is theorem-backed: the geometric claims are proved analytically and confirmed by a 29-test computational validation suite included as supplementary material.

The submission provides two mathematically distinct routes to the same result. The main letter uses convex boundedness classification of the half-space intersections. A supplementary note uses the quadratic structure of the tetrad equation together with the two-component parity structure of the spatial orthogonal group O(3). One route is convex-geometric, the other group-theoretic; they converge on the same object.

Beyond its geometric novelty, the result is relevant to the core readership of Classical and Quantum Gravity because it concerns the tetrad itself: the local frame variable required for fermion coupling and used across canonical, gauge, bimetric, and supergravity formulations. The structure reported here is kinematic rather than model-specific: it appears before any action, field equation, or quantisation choice is imposed. The submission therefore identifies a previously unreported piece of tetrad-level geometry with potential relevance across multiple existing formalisms, rather than within only one specialised framework. A classical geometric object, known for over four centuries, was present in the foundational variable of gravitational physics and went unnoticed. We believe this merits consideration as a Letter.